IBM APPLIED DATA SCIENCE

CAPSTONE PROJECT REPORT

SpaceX Falcon 9 Launch Analysis and Landing Prediction

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GitHub repository:
https://github.com/atharvv1820/IBM-Applied-Data-Science-Capstone

Project report and presentation summary

Note: This report summarizes the project work and results recorded during analysis. Confirm that every required notebook, dashboard, and visualization is included in the repository before submission.

1. Executive Summary

This project analyzes SpaceX Falcon 9 launch records to explore launch activity, payloads, orbit types, launch sites, and landing-related outcomes. The workflow covers data collection, cleaning, exploratory analysis, SQL queries, geospatial visualization, dashboard artifacts, and supervised classification.

The available structured dataset contains 90 launch records in the main analysis tables. A separate dashboard dataset contains 56 records. The observed annual launch counts in the 90-record dataset increased from 1 in 2010 to 19 in 2020. In a model comparison using a 75/25 stratified train/test split, Decision Tree and K-Nearest Neighbors each reached 82.61% test accuracy; Logistic Regression and SVM each reached 78.26%.

These metrics are preliminary: the test set contains only 23 records, so accuracy can change substantially with a different split. The target variable is the dataset's **Class** field, and it should not automatically be interpreted as mission success or landing success without checking the course's label definition.

2. Introduction

Reusable launch vehicles can reduce the cost of access to space when booster recovery and reuse are successful. This analysis uses historical Falcon 9 launch records to identify patterns associated with launch site, payload mass, orbit, flight number, and the recorded class label.

Objectives

- Collect and organize SpaceX launch data for analysis.
- Clean and transform categorical and numerical fields.
- Explore trends by year, launch site, orbit, payload mass, and outcomes.
- Use SQL to answer structured questions about the launch records.
- Visualize launch geography and prepare dashboard-style HTML outputs.
- Compare classification models and report their limitations.

3. Data Collection Methodology

3.1 API collection

The original live API attempt encountered connection/server errors (including HTTP 522/525). To keep the analysis moving, public CSV extracts were used as a fallback. This report does not claim that a successful live API pull was completed. For reproducibility, document the API endpoint, request parameters, timestamp, and response status in the notebook if the API step is rerun successfully.

3.2 Web data

A file named *spacex_web_scraped.csv* was available among the project assets. The current project notes do not establish the exact scraping code or source page used to create it. In the final submission, cite the source page and show the scraping method in the corresponding notebook if required by the rubric; do not describe a live scrape as completed unless the notebook demonstrates it.

3.3 Data assets

Asset	Observed purpose / structure
dataset_part_1.csv	90 records; launch, payload, orbit, site, outcome, booster and landing-related fields.
dataset_part_2.csv	90 records; includes Class label and fields used in SQL and model preparation.
dataset_part_3.csv	90 records; encoded categorical features including orbit, launch site, landing pad and serial.
spacex_web_scraped.csv	106 rows and 11 columns; source and scraping provenance should be documented in the notebook.

Asset	Observed purpose / structure
spacex_launch_dash.csv	56 rows and 8 columns; dashboard data including launch site, mission outcome, class and payload mass.
spacex_launch_geo.csv	56 rows and 14 columns; geographic launch records with coordinates and outcome/class fields.

4. Data Wrangling

The analysis used CSV files as inputs, inspected column names and dimensions, and prepared data for visualization, SQL and classification. The model feature set included FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, Block and ReusedCount. Categorical variables were encoded for model input. Outcome and LandingPad were excluded from the model feature set to reduce direct leakage from outcome-related fields.

Before submission, ensure the notebook explicitly shows missing-value handling, type conversion, duplicate checks, encoding, and any rows dropped or imputed. The project notes do not contain a complete log of each wrangling operation, so those steps should be verified against the notebook.

5. Exploratory Data Analysis (EDA)

5.1 Launches by year

In the 90-record analysis dataset, annual counts were: 2010: 1; 2012: 1; 2013: 3; 2014: 6; 2015: 6; 2016: 8; 2017: 18; 2018: 18; 2019: 10; 2020: 19. The record counts show a substantial increase in launch activity in later years, although the dataset's coverage and selection should be considered.

5.2 Launches by site

Launch site label	Count in dashboard CSV
CCAFS LC-40	26
KSC LC-39A	13
VAFB SLC-4E	10
CCAFS SLC-40	7

The dashboard dataset contains 56 records and uses four launch-site labels. In the separate 90-record SQL dataset, site counts were CCAFS SLC 40: 55, KSC LC 39A: 22, and VAFB SLC 4E: 13. These totals differ because the source tables cover different numbers of records and use different label formats.

5.3 Landing-related outcome rates

A derived landing indicator was created from Outcome values True Ocean, True RTLS and True ASDS. The resulting yearly rates were 0% (2010), 0% (2012), 0% (2013), 33.33% (2014), 33.33% (2015), 62.5% (2016), 83.33% (2017), 61.11% (2018), 90% (2019) and 84.21% (2020). These rates depend on the chosen outcome mapping and should be interpreted as a derived proxy, not a universal definition of launch success.

5.4 Orbit and payload patterns

The largest average payload mass by orbit in the 90-record SQL dataset was VLEO at 15,315.71 kg (14 records), followed by PO at 7,583.67 kg (9 records), and GTO at 5,011.99 kg (27 records). These figures describe this dataset only; small orbit groups, including categories with one record, are not robust for broad conclusions.

Charts prepared during analysis include Flight Number vs Launch Site, Payload Mass vs Flight Number by Site, Landing Success Rate by Orbit, Flight Number vs Orbit Type, and Payload Mass vs Orbit Type.

6. SQL Analysis

SQLite was used to query the structured CSV data. The recorded queries covered distinct launch sites, launches per site, average payload by orbit, class counts, annual launch counts, outcome distributions, landing-pad frequencies, first recorded True RTLS outcome, payload ranges for True ASDS outcomes, maximum payload, and selected outcomes during an early date interval.

Question	Result recorded
Unique launch sites	CCAFS SLC 40; KSC LC 39A; VAFB SLC 4E
Launches by site	55; 22; 13 respectively in dataset_part_2.csv
Class counts	Class 0: 30; Class 1: 60
First True RTLS record	2015-12-22; Flight 17
Maximum payload	15,600 kg at Flights 69, 74 and 77
True ASDS outcomes with 4,000–6,000 kg payload	Flights 21, 24, 29, 40, 55 and 70

SQL result interpretation should always identify the source table and any filters. Counts in this section refer to dataset_part_2.csv unless otherwise noted.

7. Interactive Map with Folium

A Folium map was generated using spacex_launch_geo.csv, containing 56 records with coordinates. The HTML output was saved as *spacex_launch_map.html*. The map supports geographic inspection of launch records and site-level patterns.

A proximity calculation based on mean coordinates gave approximately 6.90 km between CCAFS LC-40 and KSC LC-39A, 6.93 km between CCAFS SLC-40 and KSC LC-39A, and about 3,819–3,826 km between Florida-area labels and VAFB SLC-4E. The two CCAFS labels are about 0.11 km apart in the source coordinates; this may reflect alternate labels or nearby points, so do not infer separate physical facilities from the distance alone.

For the final submission, open the HTML map and verify that it renders correctly; add a screenshot to the presentation if the rubric expects visible map evidence.

8. Plotly Dash Dashboard

Dashboard-related HTML artifacts were created: dashboard_pie_summary.html and dashboard_payload_scatter.html. Additional HTML charts include landing_class_pie.html and payload_vs_class.html. The dashboard data has 56 rows; the class distribution was Class 0: 32 (57.1%) and Class 1: 24 (42.9%).

The recorded mission-outcome cross-tab showed Class 0 with 32 Success records and Class 1 with 22 Success, 1 Failure (in flight), and 1 record with unclear payload status. Therefore, the Class field should not be described as equivalent to mission success without confirming the dataset's official label definition.

A Dash app definition was created, but successful live serving was not confirmed in the project notes. If the rubric requires a working Dash app, include the source code and verify it runs, rather than relying only on static HTML exports.

9. Predictive Analysis

9.1 Setup

Four classifiers were evaluated: Logistic Regression, Decision Tree (maximum depth 4), K-Nearest Neighbors, and Support Vector Machine. The dataset was split 75% training / 25% testing with random_state=42 and stratification. The target was Class. Features included FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, Block and ReusedCount.

9.2 Model comparison

Model	Test accuracy
Decision Tree	82.61%
K-Nearest Neighbors (KNN)	82.61%
Logistic Regression	78.26%
Support Vector Machine (SVM)	78.26%

Decision Tree and KNN tied for the highest observed test accuracy. The test set contains 23 records, so the scores are preliminary and should not be treated as reliable estimates of generalization performance. A stronger evaluation would use cross-validation, report class-wise metrics, and confirm that preprocessing is fitted only on training data.

9.3 Decision Tree confusion matrix

Actual class	Predicted 0	Predicted 1
Class 0	4	4
Class 1	0	15

Recorded classification report for Decision Tree: Class 0 precision 1.00, recall 0.50, F1 0.67 (support 8); Class 1 precision 0.79, recall 1.00, F1 0.88 (support 15). The model missed half of the Class 0 test examples, which matters if identifying both classes is important.

9.4 Model conclusion

Decision Tree and KNN are tied on accuracy for this split. The Decision Tree is easier to explain, but the confusion matrix shows uneven class recall. Do not select a model on accuracy alone: compare precision, recall, F1, class balance, and validation results. Most importantly, explain what Class 0 and Class 1 mean according to the dataset documentation.

10. Conclusion and Recommendations

The analysis shows that launch activity in the available records was much higher in later years than in the early years, and that the distribution of payload mass and orbit varies across missions. SQL queries provided structured summaries; Folium supplied geographic context; Plotly outputs summarized class and payload relationships; and four classifiers were compared.

- Document the exact source and provenance for each CSV file.
- Confirm the formal meaning of the Class target before making operational claims.
- Add cross-validation and class-specific evaluation for more stable model comparisons.
- Verify that the Folium map and Dash dashboard run as expected.
- Include screenshots of key visualizations in the final presentation if required.
- Avoid broad claims from categories with very small sample sizes.

11. Creativity and Additional Insights

An additional insight from the recorded analysis is the contrast between increasing annual launch counts and the changing derived landing-outcome rate. The outcome rate is not monotonic: it rises through 2017, drops in 2018, and rises again in 2019–2020. This is a useful direction for follow-up analysis, but the chosen outcome mapping and year-wise sample sizes should be documented.

A second useful extension is a launch-site proximity analysis. The near-coincident coordinates for two CCAFS labels suggest that location names and coordinate precision should be standardized before interpreting geographic distance or treating labels as separate facilities.

12. Reproducibility and Submission Checklist

- ■ Include all required Jupyter notebooks and ensure they run from top to bottom.
- ■ Include the relevant CSV datasets and document their source.
- ■ Confirm the web-scraping notebook shows the actual method and source page.
- ■ Include SQL query code and outputs, not only summarized results.
- ■ Open spacex_launch_map.html and dashboard HTML files to verify they render.
- ■ If required, include a runnable Dash application source file.
- ■ Explain the definition of Class and the mapping used for landing-related outcomes.
- ■ Upload this PDF to the public GitHub repository and check that it opens.
- ■ Paste the repository URL into the Coursera assignment submission field.

13. Repository

GitHub repository: https://github.com/atharvv1820/IBM-Applied-Data-Science-Capstone

This report is based on the project results recorded during the analysis. It is not independent verification that every rubric requirement is complete. Confirm the required notebook outputs, source citations, and final presentation format against the current Coursera assignment instructions before submitting.